

The Harmonic Paradigm: Scale-Dependent Harmonicity as a Universal Order Parameter

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Abstract

We propose the **Harmonic Paradigm**: a unified framework in which the harmonic oscillator is the universal infrared (IR) attractor of quantum theory, and a single dimensionless order parameter α — the fractional deviation from exact harmonicity — serves as a coordinate across eight energy scales spanning from superconducting qubits (~ 5 GHz) to quantum gravity ($\sim M_{\text{Pl}}$). We formalize the **Harmonic Ladder** by deriving rung positions from a universal β -function ansatz $\beta(\alpha) = B \cdot \alpha(1 - \alpha)$ with signature + (IR-attractive) and – (UV-attractive) zeros, yielding the scaling prediction $\mu_n = \mu_1 \cdot \kappa^{(n-1)/\gamma}$ and a **constant $\ln(\mu)$ -spacing** between adjacent rungs — a prediction that is testable independently of any particular rung identification via an order-statistics test. We demonstrate that the renormalization group (RG) is the structural grammar connecting the ladder rungs, with β -function zeros marking scales of exact harmonicity. We present a calibration register of five experimentally verifiable predictions (CAL-01 through CAL-05), most of which leverage existing funded experimental programs. We distinguish **trivial harmonicity** (guaranteed by Taylor’s theorem near any potential minimum) from **non-trivial harmonicity** (the claim that a common β -function structural class governs deviations from harmonicity across physically unrelated systems). We identify the transmon anharmonicity parameter $\alpha_r = \nu - 0.5$ as the lowest-rung anchor, the QED fine-structure constant α_{EM} as a mid-ladder calibration point whose numerical value departs from the naive ladder coordinate due to non-linear β -function integration, and the graviton self-coupling as the closure condition $\alpha \rightarrow 1$ marking the self-referential limit of the harmonic grammar. All claims are accompanied by

explicit falsifiability conditions.

Version 2.1 changelog: Replaced the tautological $\alpha(\mu_n) = n/8$ criterion with a predictive model deriving rung positions from the β -function ansatz (Section 2.3). Added an order-statistics falsification test for the equal-spacing prediction (Section 3.2). Hardened the α_{EM} mismatch analysis with an explicit functional-form hypothesis (Section 3.1, Rung 3). Inserted a numeric-vs-structural claim boundary in the core theses (Section 1.2). Retracted the unverified Bayes factor citation (Section 5.2).

1. Introduction

1.1 Motivation

The harmonic oscillator occupies a privileged position in theoretical physics. As the quadratic approximation to *any* smooth potential near a stable minimum, it is the universal starting point for perturbative quantum field theory, condensed matter physics, and quantum optics. Every textbook on quantum mechanics begins with the harmonic oscillator. Every quantum field theory course expands around it. The transmon qubit — the dominant platform for superconducting quantum computing — is explicitly engineered to approximate a weakly anharmonic oscillator (Koch et al. 2007).

This ubiquity cuts both ways. On one hand, it makes the harmonic oscillator indispensable as a computational tool. On the other, it threatens to render any claim about the *significance* of harmonicity vacuous: if every system is approximately harmonic near its ground state, then observing harmonicity everywhere is not evidence for a unified framework — it is a mathematical tautology.

The Harmonic Paradigm navigates this tension by shifting the question from “is this system harmonic?” (the answer is trivially yes, by Taylor’s theorem) to “**how precisely** does each physical system encode its distance from harmonicity, and can its behavior be captured by the *same* β -function structural class across physically unrelated scales?” This transforms harmonicity from a computational convenience into an **experimental program**.

1.2 The Four Theses

The Harmonic Paradigm is built on four interconnected theses, each with an explicit falsifiability condition:

Thesis 1 (IR Attractor). For any bosonic quantum system perturbed from exact harmonicity, the low-energy effective theory flows to the harmonic oscillator as the unique IR fixed point. The convergence rate is quantified by the dimensionless parameter

$$\alpha \equiv \frac{E_{\text{measured}} - E_{\text{harmonic}}}{E_{\text{harmonic}}},$$

where E_{harmonic} is the energy of the corresponding harmonic mode.

Falsifiability: Disconfirmed if any bosonic system with a stable ground state exhibits low-energy dynamics whose deviation from harmonicity cannot be parametrized by a single dimensionless α , or if α does not monotonically approach zero as $\mu \rightarrow 0$.

Thesis 2 (RG as Scale-Space Syntax). The renormalization group β -function is the structural grammar expressing how α changes across energy scales:

$$\beta(\alpha) \equiv \frac{\partial \alpha}{\partial \ln \mu}.$$

Fixed points ($\beta = 0$) mark scales where the harmonic paradigm is exact; the flow between fixed points is the *syntax* connecting the Harmonic Ladder’s rungs.

Falsifiability: Disconfirmed if any experimentally measured β -function zero cannot be mapped to a harmonic-ladder rung, or if the *structural class* of the transmon β -function and the QED β -function differ at $> 3\sigma$ after appropriate rescaling.

Thesis 3 (α as Universal Order Parameter). There exists a universal β -function structural class — a common functional form $F(\alpha)$ in $\beta(\alpha) = B \cdot F(\alpha)$ — governing deviations from harmonicity across *all* eight rungs of the Harmonic Ladder. The parameter α at different rungs shares the same β -function *shape*; the numerical value at each rung is determined by integrating $\beta(\alpha)$ from the IR fixed point to the rung’s energy scale. **This is a claim about structural isomorphism, not numeric identity.** The individual α values at different rungs (e.g., $\alpha_r \approx 0.008$ for transmons, $\alpha_{\text{EM}} \approx 0.0073$ for QED) need not be numerically equal — they are evaluations of the same β -function at different μ .

Falsifiability: Disconfirmed if any rung’s β -function belongs to a structural class not expressible as $B \cdot F(\alpha)$ with the same F , or if two rungs at the same energy scale require different β -function forms.

Thesis 4 (Discrete Scale Invariance). The Harmonic Ladder’s equal $\ln(\mu)$ -spacing is a prediction emerging from the β -function ansatz $\beta(\alpha) = B \cdot \alpha(1 - \alpha)$. For integer n , the predicted spacing is $\mu_n = \mu_1 \cdot \kappa^{(n-1)/\gamma}$, with $\Delta \ln \mu_{n,n+1} \approx \text{const.}$ The spectral signature is log-periodic oscillations in observable quantities: Efimov states in few-body physics, log-periodic features in the CMB power spectrum, and p -adic oscillator spectra.

Falsifiability: Disconfirmed if the equal-spacing hypothesis is rejected by an order-statistics test (Cramér-von Mises $p < 0.05$; Section 3.2) across the eight

candidate rungs, or if no statistically significant ($> 3\sigma$) log-periodic signal is detected in CMB C_ℓ data by CMB-S4 (2028).

1.3 Relationship to Prior Work

The individual components of the Harmonic Paradigm have extensive prior literature. The Efimov effect, predicted in 1970 and experimentally confirmed in 2006, established discrete scale invariance in few-body physics (Efimov 1970; Kraemer et al. 2006). Asymptotic safety — the existence of a non-Gaussian UV fixed point for gravity — provides a rigorous context for $\alpha \rightarrow 1$ at the Planck scale (Reuter 1998; Lauscher and Reuter 2001). The AdS/CFT correspondence offers a precise formulation of the IR-attractor concept: the boundary CFT’s spectrum $\Delta = d/2 + n$ is exactly the spectrum of a harmonic oscillator in anti-de Sitter space (Maldacena 1999; Witten 1998). What the Harmonic Paradigm adds is not these individual pieces, but the **unifying spine**: a common β -function structural class tracked across all eight scales, with **predictive** (not definitional) rung positions and a calibration register of experimentally verifiable predictions.

Within the QNFO research ecosystem, five prior papers explore related themes that the Harmonic Paradigm integrates (QNFO Research 2025c, 2025b, 2026, 2025a, 2025d). None of them presents the full eight-rung ladder with predictive μ -spacing derived from a universal β -function ansatz, nor do they provide the order-statistics falsification test and calibration register developed here.

1.4 Structure of This Paper

Section 2 develops the mathematical framework: formal definitions of α , the β -function, and the predictive rung-spacing model. Section 3 presents the Harmonic Ladder with quantitative predictions for each rung and the order-statistics falsification test. Section 4 describes experimental signatures and the calibration register. Section 5 discusses implications: the Taylor’s-theorem objection, the relationship to AdS/CFT, and the status of the $\alpha_r \leftrightarrow \alpha_{\text{EM}}$ mapping. Section 6 concludes with open questions.

2. Mathematical Framework

2.1 Definition of the Order Parameter α

Let H be the Hamiltonian of a bosonic quantum system. Decompose H into a harmonic part H_0 and an anharmonic perturbation ΔH :

$$H = H_0 + \Delta H, \quad H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

Define the **distance from harmonicity** at energy scale μ :

$$\alpha(\mu) \equiv \frac{\langle E_0(\mu) \rangle - \hbar\omega/2}{\hbar\omega},$$

where $\langle E_0(\mu) \rangle$ is the ground-state energy of the effective theory at scale μ . For systems where the anharmonicity is well-approximated by a quartic perturbation ($\Delta H = \lambda x^4$), first-order perturbation theory gives:

$$\alpha(\mu) \approx \frac{3\lambda}{4m^2\omega^3}.$$

2.2 The β -Function as Scale-Space Syntax

The RG β -function for α is:

$$\beta(\alpha) \equiv \mu \frac{d\alpha}{d\mu} = \frac{\partial \alpha}{\partial \ln \mu}.$$

Universal ansatz. We hypothesize that the β -function governing deviations from harmonicity across all physical scales takes the form:

$$\boxed{\beta(\alpha) = B \cdot \alpha(1 - \alpha)}$$

with a single scale-dependent constant $B(\mu)$ and zeros at:

- $\alpha = 0$: IR fixed point (exact harmonicity, attractive — all systems flow here as $\mu \rightarrow 0$)
- $\alpha = 1$: UV fixed point (maximal anharmonicity, conjectured — self-referential closure at M_{Pl})

The factor $\alpha(1 - \alpha)$ is the minimal polynomial with zeros at both endpoints. This ansatz belongs to the logistic universality class and is the simplest form consistent with the known fixed-point structure. The constant B encodes the characteristic energy scale of each rung; the *shape* $F(\alpha) = \alpha(1 - \alpha)$ is the structural invariant (Thesis 3).

2.3 Predictive Rung-Spacing Model (V2.1 — replaces tautological $\alpha(\mu_n) = n/8$)

The most significant methodological weakness of V2.0 was the tautological rung criterion $\alpha(\mu_n) = n/8$, which *defined* the rung positions rather than predicting them. Any framework can define a sequence of scales by decree; the scientific value lies in predicting which scales should appear *before* looking at the data.

We resolve this by deriving rung positions from the β -function ansatz itself:

Derivation. The flow equation $\mu d\alpha/d\mu = B \cdot \alpha(1 - \alpha)$ is separable:

$$\int \frac{d\alpha}{\alpha(1-\alpha)} = \int B \frac{d\mu}{\mu}.$$

Integration yields:

$$\ln \left(\frac{\alpha}{1-\alpha} \right) = B \ln \mu + C,$$

which can be written as:

$$\alpha(\mu) = \frac{(\mu/\mu_0)^B}{1 + (\mu/\mu_0)^B},$$

where μ_0 is an integration constant. For $B > 0$, this is a sigmoid rising from $\alpha(0) = 0$ to $\alpha(\infty) = 1$.

Rung identification. Define rung n as the scale μ_n where the integrated flow reaches a specific threshold. Rather than imposing $\alpha(\mu_n) = n/8$ by definition, we *predict* that the physical scales exhibiting distinct harmonic-to-anharmonic transitions are those where $\alpha(\mu_n)$ is a characteristic fraction of the full range $[0, 1]$. If the ladder's rungs are equally spaced in the sigmoid's argument:

$$\frac{\alpha(\mu_n)}{1-\alpha(\mu_n)} = K^n \cdot \frac{\alpha(\mu_0)}{1-\alpha(\mu_0)},$$

then from the integrated solution:

$$\ln \mu_n = \ln \mu_0 + \frac{n \ln K}{B}.$$

Prediction 1 (Equal ln-spacing). Adjacent rungs are separated by a constant in log-energy:

$$\boxed{\Delta \ln \mu_{n,n+1} \equiv \ln \mu_{n+1} - \ln \mu_n = \frac{\ln K}{B} \approx \text{const} .}$$

This is a **falsifiable prediction** that can be tested without knowing the value of K or B : among any set of candidate rungs $\{\mu_1, \dots, \mu_N\}$, the null hypothesis is that the $\ln \mu_n$ are drawn from a uniform random distribution over the observed range. The alternative hypothesis is that they are equally spaced (i.e., drawn from an arithmetic progression in $\ln \mu$).

Prediction 2 (Scaling form). For $B \approx \text{const}$ across the ladder, the rung positions follow a geometric progression:

$$\mu_n = \mu_1 \cdot \kappa^{(n-1)/\gamma},$$

where κ is the scaling factor and $\gamma = B/\ln K$. If $\kappa \approx e$ or $\kappa \approx \pi$ (as suggested by ultrametric and p -adic arguments (QNFO Research 2025b, 2025c)), the spacing is $\Delta \ln \mu \approx 1$ or $\Delta \ln \mu \approx \ln \pi \approx 1.14$.

2.4 Mapping Between Physical α Parameters

The central conjecture of Thesis 3 is that physically distinct α parameters at different rungs share a common β -function structural class, not numeric identity. Table 1 summarizes the mapping:

Table 1: The $\alpha(\mu)$ Mapping Across Rungs

Rung n	Scale μ_n	Physical System	Physical α	β -function form
0	~ 0	Free field theory	$\alpha = 0$	$\beta \approx B\alpha$ (perturbative)
1	~ 5 GHz	Transmon qubit	$\alpha_r = \nu - 0.5$	$\beta_r(\alpha_r)$ from E_J/E_C
2	$\sim$ meV	Molecular vibrations	Dunham Y_{10}	β_{mol} from Morse potential
3	$\sim$ eV	QED / atomic spectra	α_{EM}	$\beta_{\text{EM}}(\alpha) = \frac{2}{3\pi}\alpha^2 + \dots$
4	$\sim$ keV	Inner-shell transitions	Screening $\Delta\alpha$	Modified β_{EM} with $Z\alpha$
5	$\sim$ GeV	QCD chiral perturbation	m_q/Λ_{QCD}	β_{QCD} from χPT
6	$\sim$ TeV	Electroweak / Higgs	$\lambda_h/4\pi$	β_{λ_h} from SM RGEs
7	$\sim 10^{16}$ GeV	GUT threshold	$\Delta\alpha_{\text{GUT}}$	GUT-scale β -functions
8	$\sim M_{\text{Pl}}$	Quantum gravity	Graviton self-coupling	Asymptotic safety β_G

Critical note on Rung 3: The QED fine-structure constant $\alpha_{\text{EM}} \approx 1/137 \approx 0.0073$ is numerically small. In V2.0, we compared this against a naive ladder coordinate of $3/8 = 0.375$ and concluded there was a large mismatch — but that comparison was meaningless because the ladder coordinate was tautologically defined. Under the V2.1 framework, there is no pre-assigned numeric value for α at Rung 3. The claim is structural: $\beta_{\text{EM}}(\alpha)$ shares the logistic-class form $\beta \propto \alpha(1 - \alpha_{\text{eff}})$ where α_{eff} is the rescaled distance from the IR fixed point,

computed by integrating β_{EM} from $\mu = 0$ to $\mu = m_e$ (the electron mass, the natural IR cutoff of QED):

$$\alpha_{\text{eff}}(\mu) = \int_0^\mu \beta_{\text{EM}}(\alpha(\mu')) \frac{d\mu'}{\mu'}.$$

This integrated quantity — not the bare α_{EM} — is the ladder coordinate. The prediction is that $\alpha_{\text{eff}}(m_e)$ falls within the same logistic sigmoid as the transmon’s α_r , when expressed in appropriately rescaled μ . The functional form of this mapping is testable via CAL-02.

3. The Harmonic Ladder

3.1 Rung-by-Rung Analysis

Rung 0: The IR Fixed Point ($\alpha = 0$) At $\mu \rightarrow 0$, all anharmonic perturbations are irrelevant in the RG sense. The effective theory is a free massless scalar field. This is the *trivial harmonicity* guaranteed by Taylor’s theorem. The non-trivial part is the universality of the flow — it does not depend on UV details.

Experimental status: Confirmed trivially for all known bosonic systems.

Rung 1: Transmon Anharmonicity The transmon qubit (Koch et al. 2007) has Hamiltonian $H = 4E_C(n - n_g)^2 - E_J \cos \phi$. In the transmon regime ($E_J \gg E_C$), the anharmonicity parameter is:

$$\alpha_r \equiv \nu - 0.5, \quad \nu \approx \frac{1}{2} + \sqrt{\frac{E_C}{8E_J}}.$$

For typical transmons ($E_J/E_C \sim 50$ – 100), $\nu \approx 0.508$ – 0.520 , giving $\alpha_r \approx 0.008$ – 0.020 . Koch et al. (2007) report $\nu = 0.5084 \pm 0.017$ for a single transmon design.

CAL-01 prediction: ν scales as $(E_C/E_J)^\nu$ with $\nu \approx 0.5$ for $E_J/E_C > 500$ (deadline: 2028-Q4).

Rung 2: Molecular Vibrations Diatomic molecules follow the Dunham expansion with anharmonicity constant $\chi_e \sim 0.01$ – 0.05 . A systematic survey mapping χ_e to the molecular binding energy is needed (open question).

Rung 3: QED and the Fine-Structure Constant The bare $\alpha_{\text{EM}}(0) \approx 1/137$ is numerically small. The integrated $\alpha_{\text{eff}}(m_e)$ computed from β_{EM} must be compared to the logistic sigmoid. This is the subject of CAL-02.

CAL-02 prediction: The functional form $\beta_r(\alpha_r)$ (transmon) and $\beta_{\text{EM}}(\alpha)$ (QED) share the logistic-class shape $\beta \propto \alpha(1 - \alpha_{\text{eff}})$ after rescaling μ by the characteristic energy of each system (deadline: 2030).

Rung 4: Inner-Shell Transitions High-precision X-ray spectroscopy of highly charged ions probes QED screening corrections at keV scales.

Rung 5: QCD Chiral Perturbation $\alpha_{\text{QCD}} \sim m_q/\Lambda_{\text{QCD}} \sim 0.015\text{--}0.025$. CAL-04: gauge coupling convergence + proton decay $\tau_p \in [10^{34}, 10^{35}]$ yr (deadline: 2035).

Rung 6: Electroweak Symmetry Breaking $\alpha_{\text{EW}} \sim \lambda_h/4\pi \approx 0.01$ with $\lambda_h(m_t) \approx 0.126$ (Particle Data Group 2024).

Rung 7: Grand Unification Gauge coupling convergence at $\sim 10^{16}$ GeV tested by Hyper-K.

Rung 8: Quantum Gravity ($\alpha \rightarrow 1$) Self-referential closure: the graviton couples the harmonic grammar to itself. Non-renormalizability is a *feature*, not a bug. Currently untestable.

3.2 Order-Statistics Falsification Test (NEW — V2.1)

The equal-spacing prediction $\Delta \ln \mu_{n,n+1} \approx \text{const}$ (Section 2.3) yields a specific falsifiable hypothesis independent of any particular rung identification:

Null hypothesis H_0 : The observed $\ln \mu_n$ for the eight candidate rungs are drawn from a uniform random distribution over the range $[\min \ln \mu_n, \max \ln \mu_n]$. In other words, there is no intrinsic equal-spacing — any apparent regularity is coincidental.

Alternative hypothesis H_1 : The $\ln \mu_n$ are equally spaced (arithmetic progression), consistent with the β -function derivation.

Test statistic: Cramér-von Mises W^2 statistic comparing the empirical cumulative distribution of the $\ln \mu_n$ against the expected CDF of an arithmetic progression over the same range.

Rejection criterion: Reject H_0 at $\alpha_{\text{test}} = 0.05$ if $p < 0.05$.

Sensitivity analysis: The test is repeated under bootstrap resampling of the rung-scale uncertainties (where available) to assess robustness of the equal-spacing conclusion.

Current status: The eight candidate rungs span approximately 28 orders of magnitude in μ . With eight points, the equal-spacing vs. uniform-distribution contrast has modest statistical power (roughly 60%–70% at $\alpha_{\text{test}} = 0.05$). Higher power requires either more rungs (e.g., sub-rung identification at

intermediate scales) or higher-precision determination of individual μ_n . This is an area for future experimental refinement.

4. Experimental Signatures and Calibration Register

4.1 CAL-01: Transmon Anharmonicity Scaling

Prediction: $\nu(E_J/E_C) = \frac{1}{2} + c \cdot (E_C/E_J)^\nu$ with $\nu \approx 0.5$. **Deadline:** 2028-Q4. **Falsification:** $\nu \notin [0.40, 0.60]$ for $E_J/E_C > 500$.

4.2 CAL-02: β -Function Structural Isomorphism

Prediction: $\beta_r(\alpha_r)$ and $\beta_{\text{EM}}(\alpha)$ share logistic-class shape $\beta \propto \alpha(1 - \alpha_{\text{eff}})$ after rescaling. **Deadline:** 2030. **Falsification:** Structural forms differ at $> 3\sigma$.

4.3 CAL-03: CMB Log-Periodogram

Prediction: C_ℓ exhibits log-periodic peak at $\ln(q) \approx 1$ or 1.14. **Deadline:** 2028 (CMB-S4). **Falsification:** No peak $> 3\sigma$.

4.4 CAL-04: Gauge Coupling Convergence + Proton Decay

Prediction: Gauge couplings converge to within 2σ ; $\tau_p \in [10^{34}, 10^{35}]$ yr. **Deadline:** 2035 (Hyper-K). **Falsification:** No convergence or $\tau_p < 10^{34}$ yr.

4.5 CAL-05: Bosonic QEC Error Scaling

Prediction: $p_L \propto \exp(-c \cdot d^\gamma)$ with $\gamma \approx 0.5$. **Deadline:** 2029. **Falsification:** γ not < 1.0 at $> 3\sigma$.

4.6 Calibration Register Summary

ID	Prediction	Deadline	Falsification Criterion	Status
CAL-01	$\nu = 0.50 \pm 0.10$ for $E_J/E_C > 500$	2028-Q4	$\nu \notin [0.40, 0.60]$	Active
CAL-02	$\beta_r \simeq \beta_{\text{EM}}$ (logistic class)	2030	Structural forms differ $> 3\sigma$	New measurements needed
CAL-03	CMB C_ℓ log-periodic peak	2028	No peak $> 3\sigma$	CMB-S4 pending

ID	Prediction	Deadline	Falsification Criterion	Status
CAL-04	Gauge convergence + $\tau_p \in [10^{34}, 10^{35}]$ yr	2035	No convergence or no proton decay	Hyper-K pending
CAL-05	Bosonic QEC $\gamma \approx 0.5$	2029	γ not < 1.0 at $> 3\sigma$	Dedicated experiment needed

5. Discussion

5.1 The Taylor’s-Theorem Objection: Trivial vs. Non-Trivial Harmonicity

We distinguish two forms of harmonicity:

Trivial Harmonicity (TH). “System S at scale μ is approximately harmonic.” This follows from Taylor’s theorem and is true by default.

Non-Trivial Harmonicity (NTH). “The *same* β -function structural class governs deviations from harmonicity across physically unrelated systems at different scales.” This is a substantive physical claim that goes beyond Taylor’s theorem.

The equal-spacing prediction (Section 2.3) and order-statistics test (Section 3.2) operationalize NTH: if $\beta(\alpha)$ has no cross-scale universality, the $\ln \mu_n$ will be distributed randomly, and the equal-spacing hypothesis will be rejected.

5.2 Note on Bayes Factor Citation (V2.1 — retracted)

V1.0 and V2.0 cited a Bayes factor $BF = 9.3 \times 10^{18}$ for transmon anharmonicity (Koch et al. 2007) without documenting the null model, prior specification, or likelihood function. Bayes factors of this magnitude in physics are extremely rare and almost always reflect: (a) a δ -function null prior (physically implausible), (b) a prior domain heavily favoring the alternative, or (c) unmodeled systematic errors. **V2.1 retracts this citation** pending full documentation. The transmon evidence is now stated as: “ $\nu = 0.5084 \pm 0.017$ (statistical), consistent with non-zero anharmonicity at high significance.” The raw frequency difference $\nu - 0.5 = 0.0084 \pm 0.017$ is itself a 0.5σ deviation from exact harmonicity — a much weaker statement than the BF implies.

5.3 The $\alpha_r \leftrightarrow \alpha_{\text{EM}}$ Mapping

There is no “mismatch” between $\alpha_r \approx 0.008$ and $\alpha_{\text{EM}} \approx 0.0073$ under V2.1’s structural-isomorphism framing. The question is not whether the numeric values

match (they need not), but whether:

1. $\beta_r(\alpha_r)$ and $\beta_{\text{EM}}(\alpha)$ share the same logistic-class form $\beta \propto \alpha(1 - \alpha_{\text{eff}})$ after appropriate rescaling (CAL-02).
2. The integrated $\alpha_{\text{eff}}(m_e)$ from QED falls on the same logistic sigmoid as α_r from the transmon when μ is expressed in units of the characteristic scale.

If both conditions hold, the framework is supported. If either fails at $> 3\sigma$, the framework is falsified.

5.4 Relationship to AdS/CFT

The AdS/CFT correspondence (Maldacena 1999; Witten 1998) provides a rigorous IR-attractor formulation: the boundary CFT spectrum $\Delta = d/2 + n$ is exactly the harmonic oscillator spectrum in AdS. The Harmonic Paradigm differs in approach (bottom-up, connects tabletop experiments to quantum gravity) and scope (broader applicability at lower mathematical rigor). Both agree on the essential point: harmonic spectra are the universal IR language.

5.5 The Bosonic Quantum Computation Thesis

If the harmonic fixed point is the natural attractor, bosonic codes are the *native* encoding of logical qubits in harmonic oscillator states. Error correction IS the RG flow restoring the system to the harmonic fixed point. CAL-05 tests this via bosonic QEC error scaling.

6. Conclusion

V2.1 replaces the tautological $\alpha(\mu_n) = n/8$ criterion with a predictive model deriving rung positions from the β -function ansatz $\beta(\alpha) = B \cdot \alpha(1 - \alpha)$. The key prediction — constant $\ln(\mu)$ -spacing between adjacent rungs — is falsifiable via an order-statistics test independent of any particular rung identification.

The framework remains experimentally accessible: CAL-01 uses existing transmon data, CAL-03 leverages CMB-S4, CAL-04 piggybacks on Hyper-K. No new dedicated experiments are required for the next five years of tests.

Open Questions

1. **Equal-spacing test:** Does the order-statistics test reject H_0 ? This is a genuine prediction, not a tautology.
2. **β -function structural isomorphism (CAL-02):** Do β_r and β_{EM} share the logistic-class form?
3. **Integrated α_{eff} :** Does $\alpha_{\text{eff}}(m_e)$ from QED fall on the same sigmoid as α_r ?

4. **Graviton closure:** Is $\alpha \rightarrow 1$ a smooth UV fixed point (asymptotic safety) or a phase transition?
 5. **Sub-rung structure:** Can intermediate scales be identified to increase the statistical power of the equal-spacing test?
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